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A machine-learning virtual source model for nanoscale transistors

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Neural networks (NNs) are powerful tools for modeling transistor characteristics from data, yet purely data-driven models are data-hungry, can yield unphysical results, and often fail to generalize. We present an integrated NN–virtual source (NN–VS) model that combines the physical rigor of the virtual source framework with the adaptive learning of neural networks. This end-to-end trainable hybrid model achieves higher accuracy, stronger data efficiency, and superior extrapolation power compared to pure NN models, while eliminating the manual parameter extraction required in the conventional VS models. Importantly, the NN–VS model demonstrates robust performance in data-limited scenarios. The model accurately captures experimental nanoscale transistor characteristics, including two-dimensional (2D) semiconductor field-effect transistors (FETs) and silicon FinFETs. The application of the model is illustrated in compute-in-memory circuit simulations with ferroelectric FETs.

As transistor channel lengths scale down to the nanometer regime and a wide variety of new channel materials are being explored, accurate and efficient transistor models that can describe a variety of nanotransistors are valuable to semiconductor design. The virtual source (VS) model¹ is a semi-empirical model based on transistor device characteristics in both subthreshold and above-threshold regimes. It has been successfully applied to a variety of nanotransistors, including silicon MOSFETs¹, carbon nanotube FETs², graphene FETs³, and GaN MOSFETs⁴. The model, however, requires manual and device-specific parameter extraction. Its parameters need to be re-extracted from device to device, which makes the process labor-intensive and operator-dependent.

Prior works have explored various data-driven approaches for semiconductor device simulation. Machine learning (ML)-based device models have recently been developed to accelerate the design loop. For instance, neural networks have been employed to directly map device geometries to current-voltage (*I*-*V*) characteristics^{5,6}, serving as high-speed surrogates for TCAD simulations. More advanced architectures have also been proposed to capture complex device behaviors, such as variability and degradation^{7–9}. These pure ML approaches demonstrate promising accuracy and speed. However, purely ML-based models can suffer from significant limitations, particularly the risk of making unphysical predictions such as non-zero source-drain current when the applied drain voltage is zero. In addition, enhancing model accuracy requires a large volume of training data, which increases the burden of precise data acquisition.

To overcome these challenges, hybrid models have been introduced. These models express the current as the product of a factor predicted by a neural network and a current predicted by a well-established transistor model, such as the Berkeley Short-channel IGFET Model (BSIM)¹⁰, a ballistic transistor model¹¹, to improve the model performance or simplified compact equations¹² to predict the drain current. The approach improves the model accuracy, but the BSIM model has hundreds of model parameters which requires significant domain knowledge and effort to extract. The ballistic transistor model involves a self-consistent loop that lowers the model speed. Recent advancements have also introduced inverse modeling frameworks for automated parameter extraction¹³ and meta-learning approaches for fast model adaptation¹⁴. While these methods significantly accelerate the modeling process compared to traditional TCAD or manual extraction, a critical limitation of these approaches is their reliance on prior data availability for the specific device under test. For instance, parameter extraction frameworks require full *I*-*V* curves to function¹³, and adaptive transfer learning models require a subset of target device measurements for fine-tuning¹⁴.

In this work, we develop a hybrid model that combines neural networks with the virtual-source approach, leveraging their strengths while overcoming the limitations for modeling a variety of nanotransistors. The NN component of the integrated model takes the device structural, material, and bias parameters as input, and produces intermediate parameters, which are the model parameters of a simplified VS model component. Notably, the device parameters branch is designed to treat physical parameters as a permutation-invariant set, inspired by the fact that structural/material parameters define the device collectively rather than in ordered tuples. This design enhances both data efficiency and generalization. The VS model component subsequently predicts device characteristics. The integrated NN–VS model framework is fully end-to-end trainable, effectively

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eliminating the need for manual parameter extraction while embedding physical insights from the VS formalism. This integrated approach significantly improves data efficiency and extrapolation capability compared to purely data-driven NN models, while also demonstrating strong generalization to unseen devices.

Approach

The integration of neural networks into a widely applicable transistor model offers a new approach to take advantage of both the strength of device physics in the transistor model and the strength of interfacing with device data of the NNs. Here, we carefully design the integrated computational model through a “divide-and-conquer” approach to accelerate and automate model parameter extraction, extend the model applicability, and improve model accuracy. While the original VS model requires manual parameter extraction for each device and pure NN models miss the device physics information, our goal is to develop an accurate, interpretable, and simple model that can be applied across multiple devices. The integrated device model we developed, as outlined in Fig. 1, makes the following contributions:

- Instead of directly predicting device behavior, our neural network predicts the intermediate parameters for a proven analytical transistor model. This approach ensures all outputs are constrained by established semiconductor physics, eliminating unphysical results and markedly improving the model’s reliability and extrapolation power.
- A permutation-invariant device branch (shared-MLP with type embeddings and pooling) yields a compact representation of heterogeneous device parameters, enhancing data efficiency and generalization to unseen designs.
- The model’s real-world effectiveness is demonstrated across diverse technologies. The model accurately captures experimental data from both industrially-dominant silicon FinFETs and next-generation 2D-semiconductor FETs, proving its broad applicability from today’s technology to tomorrow’s.

Analytical transistor theory

The analytical transistor model is based on the virtual source (VS) theory. The VS model can describe transistor characteristics ranging from drift-diffusion and velocity saturation to quasi-ballistic and ballistic transport regimes^{15,16}. As described extensively in references^{15,16}, it captures characteristics in different transport regimes by assigning model parameters with proper dependence on the transistor’s physical and bias conditions. By combining the VS model with a neural network (NN) in an integrated framework, the NN component provides the flexibility to capture parameter dependencies and account for non-ideal effects such as temperature.

The analytical transistor model component in the integrated model framework takes the output from neural networks and outputs the device characteristics. It is designed to satisfy the following requirements: First, as a step in the overall computational graph that links the device structural, material, and bias parameters to device characteristics, the analytical model needs to be differentiable. Second, the model needs to capture the empirical device characteristics, to alleviate training difficulty and data demand by integrating the domain knowledge of transistor physics. To ensure model trainability while incorporating physical insights, we adopt a simplified phenomenological expression for the charge at the virtual source:

$$Q_i = C_g n \phi_t \ln \left(1 + \exp \left(\frac{V_G - V_t}{n \phi_t} \right) \right), \quad (1)$$

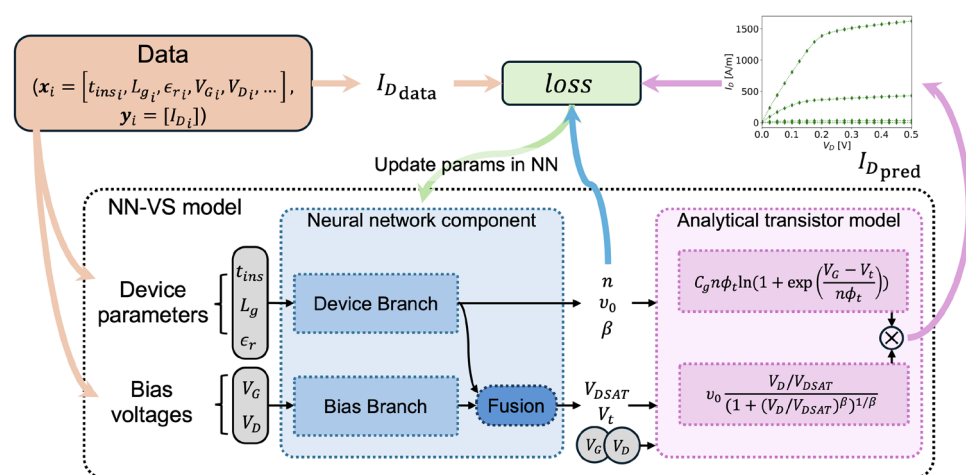


Fig. 1. Schematic overview of the integrated neural network-virtual source (NN-VS) model framework. The black dashed-line box encloses the model’s computational graph, where neural networks (NNs) predict intermediate physical parameters. These parameters (n, v_0, β, V_{DSAT} , and V_t), visually depicted connecting the NN stage to the analytical model stage in the schematic, are then used as inputs for a differentiable analytical transistor model based on virtual-source theory.

where C_g is the gate capacitance, which is the serial combination of the gate insulator capacitance and the semiconductor or quantum capacitance, $1/C_g = 1/C_{ins} + 1/C_Q$. The parameter n is a unitless parameter that determines the subthreshold swing, $\phi_t = k_B T/q$ is the thermal voltage, V_G is the applied gate voltage, and V_t can be regarded as the threshold voltage, which shifts the entire I_D - V_G characteristics along the V_G axis. The gate insulator capacitance is given by $C_{ins} \approx n_g \epsilon_0 \epsilon_r / t_{ins}$, where t_{ins} and ϵ_r denote the gate insulator thickness and relative dielectric constant, respectively. Here, $n_g = 1$ for a top or bottom-gated device, and $n_g = 2$ for a double-gated transistor or FINFET. For a practical transistor structure, the gate capacitance is often dominated by the gate insulator capacitance and slightly degraded by the quantum capacitance, which relaxes accuracy requirements for calculating the semiconductor quantum capacitance. We estimate the quantum capacitance by using the effective mass approximation for the lowest semiconductor subband with spin and valley degeneracy. For example, for a top-gated 2D semiconductor FET with a high- κ dielectric constant of 20 and insulator thickness of $t_{ins} = 3$ nm and a typical transition metal dichalcogenide channel material, the quantum capacitance is about one order of magnitude larger than the gate insulator capacitance. The value of n , which is essentially the ratio between the subthreshold swing SS and the ideal limit, $SS_0 = \phi_t \ln(10)$, depends on the transistor device parameters. The value of the threshold voltage, V_t , depends on both the transistor device parameters and the applied bias due to effects such as drain-induced barrier lowering (DIBL). As shown in Fig. 1, these parameters are determined by the output from the neural networks, which take device parameters as input, as described later.

The carrier injection velocity is expressed as

$$v_i = v_0 F, \quad (2)$$

where v_0 is a bias-independent velocity value determined by a neural network from device parameters, and the unitless factor F depends on the bias as:

$$F = \frac{V_D/V_{DSAT}}{(1 + (V_D/V_{DSAT})^\beta)^{\frac{1}{\beta}}}, \quad (3)$$

where β is a unitless exponential coefficient¹ and V_{DSAT} is the drain saturation voltage determined by the neural network output from device parameters. The source-drain current per transistor width W is expressed as the product of the carrier density and injection velocity,

$$\frac{I_D}{W} = Q_i v_i. \quad (4)$$

The above analytical model is simple, but when combined with the NN model component, it is sufficient to accurately describe both the advanced simulation and experimental data of various nanoscale transistors, as discussed later.

Neural network design

The neural network component, as depicted in Fig. 1, is designed with a dual-branch architecture. This structure is physically motivated to disentangle the influence of a transistor's static, intrinsic properties from its dynamic operational state under bias.

The first branch, the device branch, processes intrinsic device parameters such as gate insulator thickness (t_{ins}), gate length (L_g), and dielectric constant (ϵ_r). These parameters define the fundamental physical structure of the transistor. To capture the fact that device parameters form an unordered set of heterogeneous physical quantities rather than an ordered vector, the device branch employs a shared-MLP with learnable type embeddings, followed by permutation-invariant pooling. Each parameter (e.g., t_{ins} , L_g , ϵ_r) is tokenized with its type embedding and passed through the same MLP, enforcing a universal mapping conditioned on the parameter type. The outputs are aggregated via mean pooling to form a compact device representation that is invariant to input order and more data-efficient than treating parameters as a simple concatenated vector. This representation is then directly used to determine parameters that are largely governed by the device's structure, such as the subthreshold swing factor n , the injection velocity v_0 , the saturation voltage V_{DSAT} , and the fitting parameter β .

The second, parallel bias branch, processes the external operational voltages, primarily the gate voltage (V_G) and drain voltage (V_D). It is designed to capture the transistor's general response to electrical stimuli.

A key design choice is late fusion via a modulation layer¹⁷, where intrinsic device properties (device branch) modulate the response to external biases (bias branch). Concretely, given the bias-branch activations h , the fusion applies a feature-wise affine transform:

$$\text{Fusion}(h \mid \rho, \omega) = \rho \odot h + \omega, \quad (5)$$

where $\odot$ denotes element-wise multiplication and ρ, ω are feature-wise scale and shift vectors predicted by the device branch. This implements the physical notion that device structure scales and shifts the bias response before the analytical VS stage.

Finally, the modulated feature vector is used to predict only the threshold voltage V_t . This is because V_t is uniquely dependent on both the intrinsic device structure and the applied biases (e.g., through effects like DIBL). Therefore, it is most accurately predicted from the fused representation that combines information from both domains.

It is worth highlighting that this physics-aware disentanglement enables high accuracy using only lightweight, shallow MLPs for each component, avoiding the need for over-parameterized black-box models. The specific dimensions (typically 2–3 layers), along with the hyperparameter optimization methodology (grid search on validation set), are provided in Supplementary Information.

Data generation and model training

Data generation: To illustrate the application of the integrated model to nanoscale transistors, we apply it to both nanotransistors based on two-dimensional (2D) semiconductors and silicon FinFETs.

The 2D MOSFET has a double gated structure similar to those in Ref.¹⁸. The channel material is assumed to be monolayer MoS₂. The device I - V and Q - V characteristics of the 2D FET are obtained from numerical device simulations based on self-consistent solutions to the non-equilibrium Green's function (NEGF) quantum transport equation and the electrostatic Poisson equation. Phonon scattering is treated using the self-consistent Born approximation in the NEGF simulation^{18,19}. The simulation calculates the intrinsic device characteristics without parasitic contact resistance, and has been previously applied to assess 2D FETs near the scaling limits¹⁸.

Furthermore, to illustrate that the integrated NN-VS model is general and can be applied to other device data, we also apply it to the device data of the silicon FinFET. The FinFET device data is generated by the Berkeley short-channel IGFET model for common multi-gate transistors (BSIM-CMG) model²⁰.

Training: The integrated NN-VS model is designed to be end-to-end differentiable, enabling direct training using standard machine learning libraries. We employ the Adam optimizer with a learning rate of 0.001 and a batch size of 128 for training. A custom loss function is formulated to balance multiple objectives critical for accurate transistor modeling:

$$\text{loss} = \alpha \cdot \text{MSE}(I_D) + \eta \cdot \text{MSE}(\log(I_D)) + \gamma \cdot \text{MSE}(n) + \delta \cdot \text{MSE}(v_0), \quad (6)$$

where α, η, γ , and δ are the coefficients for each loss term. In this work, we set $\alpha = 1, \eta = 1, \gamma = 0.1$ and $\delta = 10^{-4}$. The coefficient δ is set to a relatively small value because the ground truth for injection velocity v_0 is extracted with inherent uncertainty; thus, this term serves as a soft regularization constraint rather than a hard target. The first two terms aim to minimize the mean squared error on both the linear and logarithmic scales of the drain current I_D . This dual objective ensures accuracy from the low-current subthreshold region to the high-current above-threshold region. The loss function also includes terms for the intermediate physical parameters: $\text{MSE}(n)$ encourages accurate prediction of the subthreshold swing factor, and $\text{MSE}(v_0)$ provides a constraint on the injection velocity.

The dataset used for training and evaluation comprises simulation results for 2D FETs, obtained from NEGF simulations as described above, with varying device parameters. The gate insulator thickness t_{ins} ranges from 1.0 nm to 3.5 nm in 0.5 nm increments, the gate length L_g varies from 10 nm to 30 nm in 2 nm steps, and the relative dielectric constant ϵ_r takes values of 3.9, 10, 15, 20, and 25. All models, including the proposed NN-VS model and the baseline comparison, were trained using data with gate voltages V_G swept from 0 V to 0.5 V in 0.0125 V steps and drain voltages V_D swept from 0.001 V to 0.501 V in 0.0125 V steps. The model formulation inherently guarantees zero current at $V_D = 0$, so this specific bias point was excluded from the training data. The dataset was partitioned such that 70% of the devices were allocated for training, while the remaining 30% were reserved for testing. It is important to note that this data splitting strategy represents a more challenging generalization task compared to approaches where data points from the same device might appear in both training and testing sets. Here, the test set consists of devices entirely unseen by the model during training, providing a robust assessment of the model's ability to generalize to new device geometries and materials. All models discussed subsequently, including the baseline models, were trained for 500 epochs, and the results presented correspond to the epoch exhibiting the best performance on the unseen test set.

Baseline NN model for comparison

To rigorously evaluate the performance, accuracy, and data efficiency of the proposed physics-informed NN-VS model, we established a baseline model for comparison. The baseline model was trained and evaluated on the exact same dataset generated by the NEGF and BSIM-CMG simulations described in the **Data Generation** section.

The baseline model is a state-of-the-art multi-layer perceptron (MLP) model²¹, an architecture that has been successfully applied to transistor modeling^{8,9,22,23}.

Specifically, following the methodology in Ref.²¹, to enhance numerical stability across subthreshold and strong inversion regions, the baseline predicts a transformed variable $y_1 = \ln(I_D/V_D)$, such that $I_D = V_D \cdot e^{y_1}$. Furthermore, the training is guided by a composite physics-aware loss function that includes penalties not only for the current error but also for its first and second-order derivatives (conductance g_m, g_d and their derivatives g'_m, g'_d). This ensures that the MLP learns smooth, physically plausible characteristics. The MLP employed consists of 4 hidden layers, each with 256 neurons.

All other model training settings, including the optimizer parameters of the Adam optimizer with a learning rate of 0.001, a batch size of 128, the training data split, and the training duration of 500 epochs were kept identical to those used for the NN-VS model to ensure a fair comparison.

Results and discussions

Model performance assessment on accuracy and extrapolation power

We first compare the model accuracy of the proposed NN-VS model to the baseline MLP model. Figure 2 compares the I - V characteristics predicted by the NN-VS model and the baseline MLP model against the

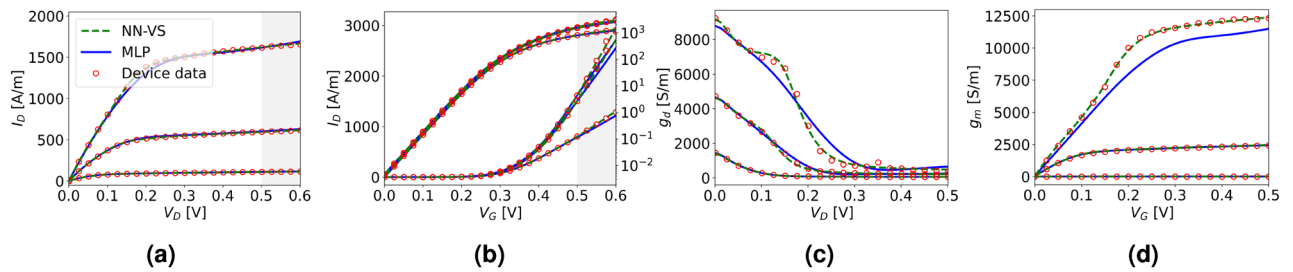


Fig. 2. Comparison of the models (lines) with the NEGF simulation data (symbols) for a monolayer MoS₂ FET with semi-infinite doped source and drain extensions. The double-gated FET has a gate length of $L_g = 10$ nm, gate insulator thickness of $t_{ins} = 2.5$ nm and dielectric constant of $\epsilon_r = 10$. The green dashed line is for the NN-VS model. The blue solid line is for the baseline MLP model (a) I_D - V_D characteristics for $V_G = 0.3, 0.4$, and 0.5 V. (b) I_D - V_G characteristics for $V_D = 0.101, 0.301$, and 0.501 V. The gray shaded regions indicate extrapolation beyond the training bias range. The NN-VS model (green dotted lines) and the baseline MLP model (blue lines with circular markers) are compared to the NEGF data (red circles). (c) Drain conductance $g_d = \partial I_D / \partial V_D$ versus V_D for the same V_G values as in (a), and (d) Transconductance $g_m = \partial I_D / \partial V_G$ versus V_G for the same V_D values as in (b), within the interpolation bias region.

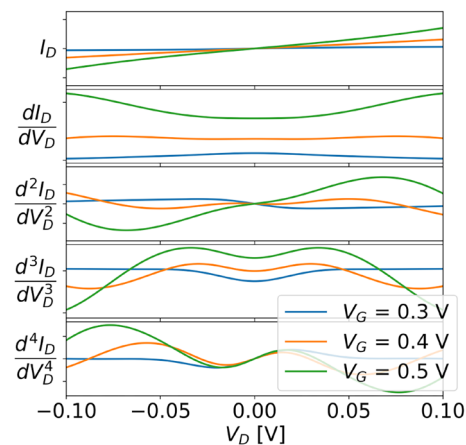


Fig. 3. Gummel Symmetry Test are shown up to the fourth derivative, the model employs a symmetric mirroring approach for the negative drain bias region. The results indicate strict physical continuity and smoothness of higher-order derivatives at the zero-crossing point.

reference NEGF data for a representative test device with a geometrical configuration (L_g, t_{ins}, ϵ_r) not included in the training set.

As illustrated in Fig. 2a and b, both the NN-VS and baseline MLP models can achieve good agreement with the NEGF I_D - V_D and I_D - V_G characteristics within the interpolation region. However, a more stringent assessment of model fidelity comes from examining their first-order derivatives - the drain conductance g_d and transconductance g_m - which are numerically calculated from the models' predicted I - V curves. These derivatives are presented in Fig. 2c and d. In this regard, the NN-VS model demonstrates a clear advantage, accurately capturing these derivative features within the interpolation region. This is particularly noteworthy as it is achieved without any explicit gradient-matching terms in its loss function, indicating a genuine learning of the underlying device characteristics. The MLP, while reasonable for I - V interpolation, typically shows larger deviations in capturing these crucial derivative behaviors. Beyond the interpolation region, the NN-VS model also shows high accuracy for extrapolation, as shown by the gray shaded regions in Fig. 2(a-b). While the baseline MLP model exhibits significant errors when extrapolating to unseen V_G bias conditions, the NN-VS model maintains high accuracy.

To further strictly validate the model's physical consistency, we performed the Gummel Symmetry Test (GST). As shown in Fig. 3, the GST plot remains continuous and smooth through the fourth derivative. This capability is intrinsic to combining the neural network with the analytical virtual source theory, the model inherits the inherent smoothness and physical symmetry of the analytical equations.

Computational efficiency

In addition to predictive accuracy, the proposed model offers a transformative advantage in computational speed. The reference NEGF simulations involve iteratively solving the coupled quantum transport and

electrostatic equations, a computationally intensive process that can take minutes per bias point. In contrast, the NN-VS model generates results via a single forward pass through lightweight MLPs and analytical formulas. Consequently, the inference speed of the NN-VS model is estimated to be over 9×10^6 times faster than the NEGF device simulations. Furthermore, due to the lightweight architecture of each component, the inference time of the NN-VS model is comparable to that of simple black-box MLP baselines, demonstrating that the integration of physical knowledge imposes negligible computational overhead while significantly enhancing model reliability.

Training data efficiency

We evaluated training data efficiency by reducing the training dataset to a coarser grid with a step size of 0.0375 V. Although this voltage step might suffice for interpolating bias points on a known device, the interpolation approach cannot extrapolate beyond the training bias range or predict the characteristics of transistors with unseen geometrical parameters.

Figure 4 compares the NN-VS and baseline MLP model performance under this sparse data condition. While the baseline MLP can maintain a semblance of accuracy within the training data range (interpolation), its predictive power quickly deteriorates outside this range. Specifically, significant errors emerge when extrapolating the I_D - V_G characteristics. In contrast, the NN-VS model maintains excellent alignment with NEGF data in both interpolation and extrapolation, demonstrating its superior robustness and generalization capability.

Training by dataset with multiple devices and interpretability of intermediate model parameters

In the original virtual source model, the parameters are extracted for an individual device, and it needs to be re-parameterized again if the device's geometric or material parameters change¹. By harvesting the ML capabilities in the NN-VS model, the model is end-to-end trainable by ML techniques based on the data encompassing a set of devices with varying structural parameters, such as gate insulator thickness and gate length. The NN learns a generalized relationship across multiple devices with varying device parameters, which automates and removes human bias in parameter extraction. This learned relationship maps device structural and material parameters to the intermediate physical parameters, which are the inputs to the VS model component for an entire class of devices. Analyzing this input-output relationship embedded within the trained NN can therefore yield broader physical insights than single-device extraction methods.

The intermediate parameters of the proposed NN-VS model, which serve as the output of the NN component and simultaneously as input to the analytical model, as shown in Fig. 1, can offer valuable insights into device characteristics. As an example, the NN part of the hybrid model predicts n , which is the ratio between the subthreshold swing to its ideal limit, $n = SS/SS_0$, where $SS_0 \approx 60$ mV/dec at room temperature. SS values are calculated from this model-predicted n . Figure 5 compares this derived SS against NEGF-extracted SS_{Data} , demonstrating high predictive accuracy across variations in L_g , t_{ins} , and two dielectric constants, which confirms the accurate learning of n .

This capability allows prediction of the physically meaningful VS model parameters directly from the device structural and material parameters by using the trained NN component. It also offers the flexibility to optimize the design of the device characteristics, such as the subthreshold swing, by using the NN model component.

Model validation by the silicon FinFET data

To further demonstrate its versatility and its capability to capture environmental variations, the NN-VS model was applied to silicon FinFETs using I - V characteristics generated by the industry-standard BSIM-CMG model²⁰. The dataset encompasses a wide temperature range, sweeping from -50°C to 125°C in 25°C steps.

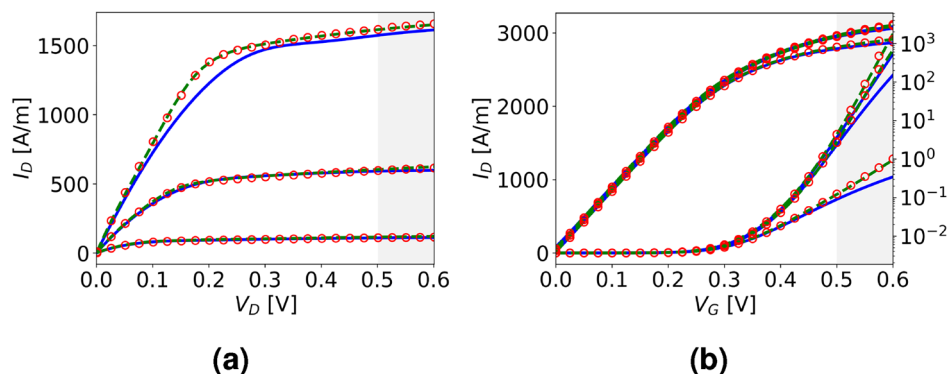


Fig. 4. Training data efficiency comparison for the NN-VS and baseline MLP when trained on a reduced dataset. The training data was reduced by using a 0.0375 V step size for both V_D and V_G sweeps, resulting in a 14×14 grid per device, which is approximately 11.7% of the original data size on a 41×41 bias grid. Both models were trained using this sparse dataset following the previous pipeline, and their predictions for the same modeled device in Fig. 2 are shown for (a) I_D - V_D and (b) I_D - V_G characteristics.

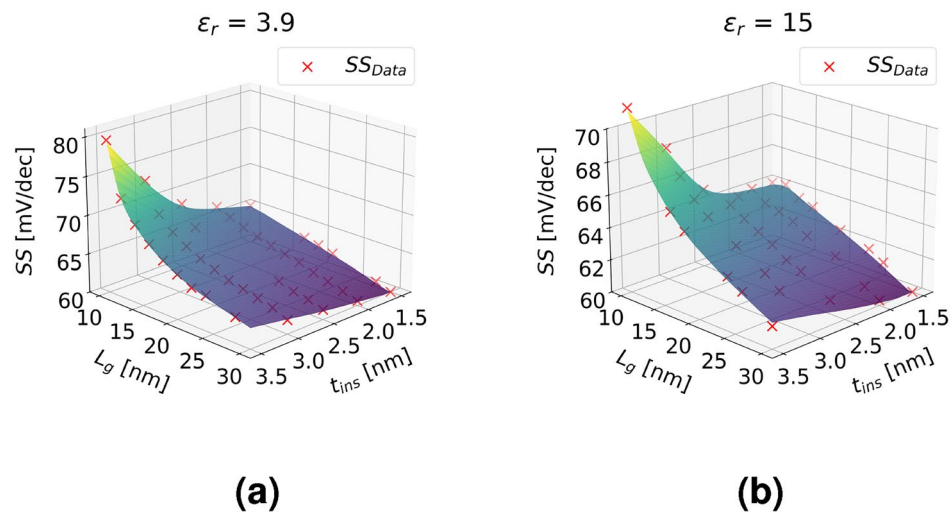


Fig. 5. Predicted subthreshold swing (SS) surfaces as a function of gate length (L_g) and insulator thickness (t_{ins}) for 2D FETs with (a) relative dielectric constant $\epsilon_r = 3.9$ and (b) $\epsilon_r = 15$. The surfaces represent the NN-VS model predictions, while the red crosses (‘x’) indicate values extracted directly from the NEGF simulation data. The device is a monolayer MoS₂ with other parameters the same as in Fig. 2.

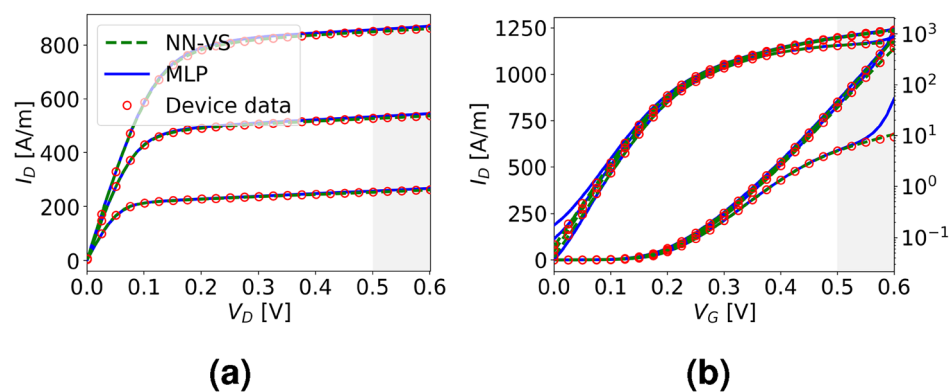


Fig. 6. Comparison of the MLP and NN-VS model predictions (lines) against the BSIM-CMG reference data (symbols) for an n-type FinFET with a single fin. The device parameters are: gate length $L_g = 10$ nm, fin thickness 5 nm, gate-oxide thickness 2.5 nm, $\epsilon_r = 20$, temperature 25°C and a metal-gate work function of 4.3 eV; all other parameters adhere to BSIM-CMG defaults. (a) I_D - V_D characteristics are shown for the same V_G values as Fig. 2a. (b) I_D - V_G characteristics are depicted for the same V_D values as Fig. 2b.

To incorporate temperature effects, the operating temperature is added as an input to the device branch of the neural network, and the thermal voltage $\phi_t = k_B T / q$ in the analytical component is dynamically updated based on the physical temperature T . The model's robustness was validated by training on a sparse dataset with 0.0375 V steps and testing on a dense dataset with 0.0125 V steps. Additionally, the test devices possessed geometries that were excluded from the training phase.

Figure 6 presents the I - V predictions. The NN-VS model accurately captures the FinFET characteristics, not only within the training bias region but also extending robustly into the extrapolation regions. This strong performance, particularly in V_G extrapolation where purely data-driven models like the baseline MLP struggle. These results with FinFETs confirm the broad applicability and effectiveness of the NN-VS framework across diverse nanotransistor architectures and operating conditions.

Model validation by the experimental data

Next, we show that the model can be applied to effectively model the experimental data. Two types of devices, a silicon nanoribbon FET²⁴ and a InSe 2D-semiconductor FET²⁵ are modeled.

Figure 7 shows the model's validation against experimental data for a silicon gate-all-around nanoribbon transistor with a gate length of $L_g = 75$ nm. The NN-VS model accurately reproduces both the I_D - V_D output and I_D - V_G transfer characteristics. The model's predictions achieve excellent fidelity with the measured data, attaining an R^2 value greater than 99.99%.

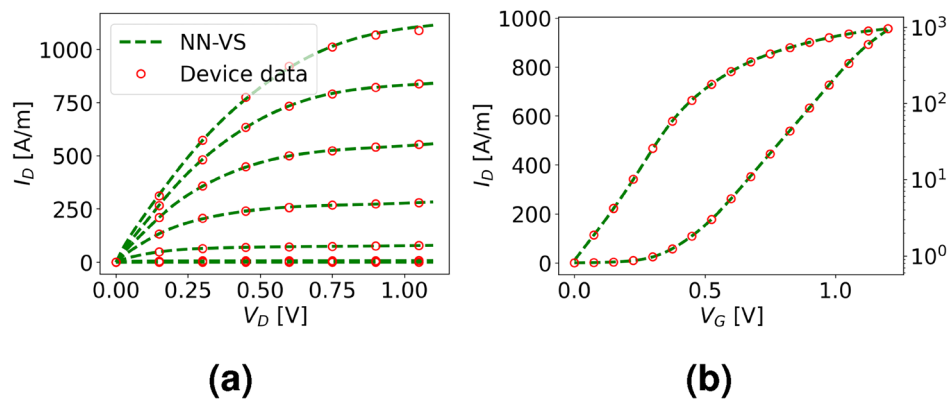


Fig. 7. Comparison of NN-VS model predictions (lines) and experimental device data (symbols)²⁴ of silicon gate-all-around nanoribbon transistors at $L_g = 75$ nm (a) I_D - V_D characteristics at $V_G = 0$ V to 1.2 V, step = 0.2 V (b) I_D - V_G characteristics at $V_D = 0.65$ V on log scale (left) and linear scale (right).

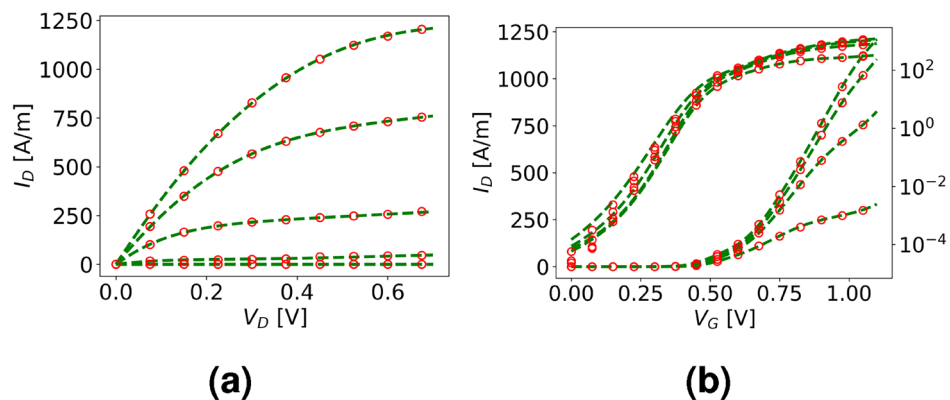


Fig. 8. Comparison of NN-VS model predictions (lines) and experimental device data (symbols)²⁵ of an InSe FET at $L_g = 10$ nm (a) I_D - V_D characteristics at $V_G = 0.1$ V to 1.1 V, step = 0.2 V (b) I_D - V_G characteristics at $V_D = 0.1$ V to 0.7 V with 0.2 V/step on log scale (left) and linear scale (right).

Figure 8 further demonstrates the model's versatility using experimental data from a nanoscale InSe FET with a gate length of $L_g = 10$ nm. The model predictions again show outstanding agreement across all bias conditions, including the subthreshold region, resulting in an R^2 value exceeding 99.99%. This successful validation on different device architectures and materials underscores the broad applicability and high accuracy of the NN-VS framework for experimentally demonstrated devices.

Model application to circuit simulation

To demonstrate the application of the NN-VS source model in circuit simulations, we simulated a crossbar array (CBA) circuit for analog compute-in-memory (CiM) multiply-accumulate (MAC) computing functionalities, as shown in Fig. 9. The MAC computation is crucial for implementing neuromorphic computing systems, as well as for solving linear algebra computation problems²⁶. The device component in the CBA is a ferroelectric FET (FeFET)^{26,27}. The device can be modeled as a ferroelectric capacitor in series with the gate electrode of a MOSFET, following Ref.²⁸, as illustrated in Fig. 9a and b. The ferroelectric layer is modeled with a thickness of $t_{FE} = 10$ nm and a relative dielectric constant of $\epsilon_{r,FE} = 25$. The transistor I - V characteristics are computed by the proposed NN-VS model. The internal node voltage V_{MOS} , as shown in Fig. 9b, is computed by solving the charge conservation relation at the internal node as described in detail in Supplementary Information. We focus on demonstrating the use of the transistor model in circuit-level applications. The FeFET devices operate in read mode without polarization switching, and the polarization states are held at fixed, pre-programmed levels to represent matrix elements in multiply-accumulate (MAC) operation.

Based on the model, we simulate a 4×4 FeFET CBA circuit as shown in Fig. 9c. We focus on the read operation in which the ferroelectric polarization values are fixed. The MAC operation is performed by using a scheme as demonstrated experimentally in Ref.²⁶. Each multiplier can take four values. The ferroelectric polarization encodes 4 different polarization values, $P_{0'}$, $P_{1'}$, $P_{2'}$, $P_{3'}$, which control the threshold voltage and result in 4 approximately equally spaced current values at $V_G = V_D = V_{DD}$, as illustrated in Fig. 9d. To perform the matrix-vector multiplication, the matrix elements are stored into the ferroelectric states of the

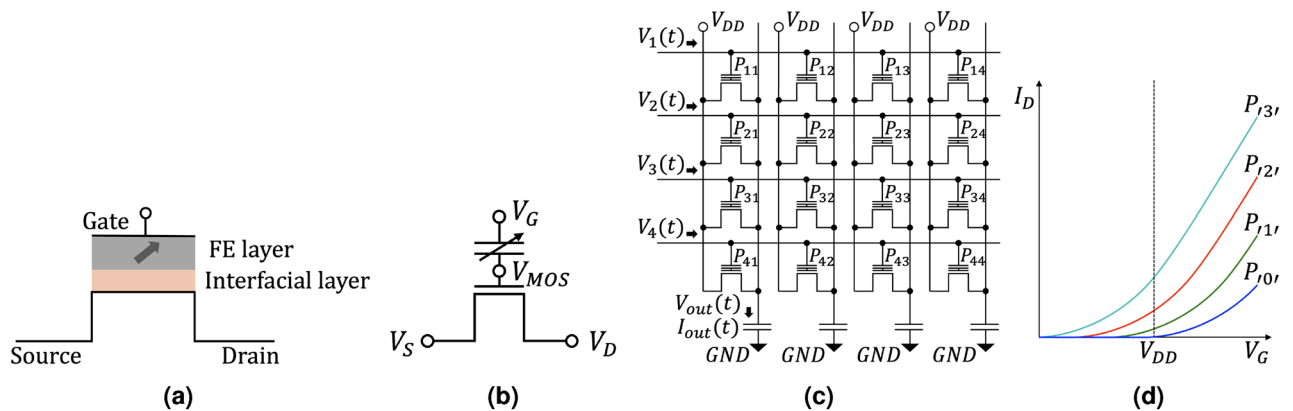


Fig. 9. (a) Schematic of a FeFET consisting of a gate stack with an interfacial dielectric layer (IL) and a ferroelectric (FE) layer on top of the transistor channel. (b) Circuit-level representation of the FeFET as an FE capacitor in series with a MOSFET, where V_{MOS} is the effective MOS gate voltage determined by the IL. The applied gate voltage V_G drops partially across the FE ($V_{FE} = V_G - V_{MOS}$). (c) A crossbar array using FeFET devices at the array intersections to perform multiply-accumulate (MAC) operations. (d) Schematic I_D - V_G characteristics of the transistor with four different FE polarization values, resulting in four states with different threshold voltages.

FeFETs in the CBA, and the vector is encoded into the width of the applied voltage pulse in each row. The MAC computation result is read out as the voltages on the load capacitors of the columns of the CBA.

To obtain the ground truth for the transistor I-V and Q-V characteristics, we established a high-fidelity benchmark using a two-step method: 1) An extensive NEGF dataset, detailing the I_D and Q_G characteristics of the FeFET's underlying MOSFET component, was created by sweeping both V_G and V_D from 0 to 0.6 V on a 97×97 grid. 2) Further increasing the dataset density results in a negligibly small impact on the simulation results compared to the relative error scale we are interested in for assessing the model accuracy. We interpolate this high-density data in circuit simulations to generate ground truth for the circuit simulation results, to benchmark the accuracy of the trained model.

As shown in Fig. 10, the transient circuit simulation is performed by a time-stepping scheme that numerically solves the charge conservation equation to update the internal gate potential at each step, and it outputs time-dependent current and voltage values. Because the load capacitor collects the current flow through all transistors in a column, its final charge is determined by the sum of the product of the pulse width and the average current through a FeFET over all rows. The results show that the transient circuit characteristics, including both the stepwise current response and the gradual capacitor voltage accumulation, can be accurately reproduced by the NN-VS model. Across all input pulse combinations, the NN-VS predictions closely track the reference device-level NEGF data in both $I_{out}(t)$ and $V_{out}(t)$, demonstrating the model's fidelity in capturing temporal charge integration effects and current modulation induced by FeFET polarization states.

Conclusion

In this work, we introduced the NN-virtual source (NN-VS) model, a physics-informed framework that integrates the analytical rigor of the virtual source model with the adaptability of neural networks. By embedding physical constraints into a data-driven architecture, the NN-VS model achieves both high accuracy and strong generalization, even under sparse training conditions. This end-to-end trainable framework not only eliminates manual parameter extraction but also extends seamlessly across technologies, from 2D FETs to silicon FinFETs, and to circuit-level applications. The model accurately describes experimental device characteristics of industrial silicon FETs and emerging 2D-semiconductor-based FETs. These results highlight a scalable path toward next-generation device models that are accurate, data-efficient, and physically interpretable.

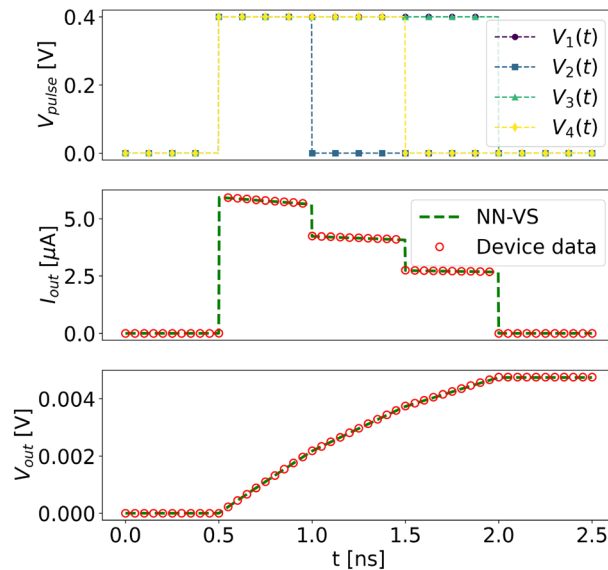


Fig. 10. Results of transient simulation of the first column of the 4×4 FeFET CBA as shown in Fig. 9c. The top panel shows the four input pulses (V_1 , V_2 , V_3 , V_4) with widths of 1.5, 0.5, 1.5, and 1 ns, which corresponds to the pulse width encoding of '3', '1', '3', and '2', respectively. The pulse amplitude is 0.5 V. The middle and bottom panels plot the output current $I_{out}(t)$ and voltage $V_{out}(t)$ on a load capacitor. The FeFET as schematically shown in Fig. 9a has an interfacial layer thickness of 1 nm and a relative dielectric constant of $\epsilon_{r,IL}=10$ with a 10 nm-thick ferroelectric layer, and a 30 nm channel width. The gate length is $L_g = 10$ nm and the load capacitance is $C_L = 1.2$ pF. The four polarization states encoding '0', '1', '2', and '3' as shown in Fig. 9d results in $I_D \approx 0.0 \mu\text{A}$, $1.5 \mu\text{A}$, $3.0 \mu\text{A}$, $4.5 \mu\text{A}$, respectively, with $P_{11} = P_{2'}$, $P_{21} = P_{1'}$, $P_{31} = P_{0'}$, and $P_{41} = P_{2'}$. The lines show the circuit simulation results with the NN-VS model, and the symbols show the simulation results with the device NEGF data.

Data availability

The codes developed in this work, including the NN-virtual source (NN-VS) model and the baseline model, are available at <https://github.com/UF-QuantumNanoLab/NN-VS>

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Author contributions

Q.Y. designed the model, carried out the simulations, performed data collection and analysis, and wrote the manuscript. J.G. proposed the initial concept, supervised the research, and contributed to discussions and manuscript editing. Both authors reviewed and approved the final version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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